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## **HDOC Day-10 – C Programming**

### **Question-**

Write a **menu-driven C program** using **switch-case** to input a positive integer and perform one of the following operations:

1. Find the **sum of squares of its digits** using a `while` loop.
  2. Find the **sum of cubes of its digits** using a `for` loop.
  3. Find the **difference between the sum of cubes and sum of squares** of its digits using a `do-while` loop.
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### **1. Algorithm**

**Step 1:** Start the program.

**Step 2:** Declare integer variables `num`, `temp`, `digit`, `choice`, `sumSquare`, `sumCube`, and `difference`.

**Step 3:** Input a positive integer `num`.

**Step 4:** Display the following menu:

- 1 → Sum of squares of digits
- 2 → Sum of cubes of digits
- 3 → Difference between sum of cubes and sum of squares

**Step 5:** Input the user's `choice`.

**Step 6:** Use a `switch` statement to perform the selected operation.

#### **Case 1: Sum of Squares Using `while`**

1. Set `temp = num` and `sumSquare = 0`.

2. Repeat while `temp > 0`.
3. Extract the last digit using `digit = temp % 10`.
4. Calculate `sumSquare = sumSquare + digit * digit`.
5. Remove the last digit using `temp = temp / 10`.
6. Display `sumSquare`.

### Case 2: Sum of Cubes Using `for`

1. Set `sumCube = 0`.
2. Use a `for` loop with `temp = num`.
3. Extract each digit using `digit = temp % 10`.
4. Calculate `sumCube = sumCube + digit * digit * digit`.
5. Remove the last digit using `temp = temp / 10`.
6. Display `sumCube`.

### Case 3: Difference Using `do-while`

1. Set `temp = num`, `sumSquare = 0`, and `sumCube = 0`.
2. Execute the `do-while` loop.
3. Extract the last digit using `digit = temp % 10`.
4. Add its square to `sumSquare`.
5. Add its cube to `sumCube`.
6. Remove the last digit using `temp = temp / 10`.
7. Continue until all digits are processed.
8. Calculate:

`difference = sumCube - sumSquare`

9. Display the difference.

**Step 7:** If the choice is not 1, 2, or 3, display "**Invalid choice**".

**Step 8:** Stop.

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## 2. Test Case Table

| Test Case | Number | Choice | Operation      | Expected Output | Actual Output | Result |
|-----------|--------|--------|----------------|-----------------|---------------|--------|
| 1         | 123    | 1      | Sum of squares | 14              | 14            | Pass   |
| 2         | 123    | 2      | Sum of cubes   | 36              | 36            | Pass   |
| 3         | 123    | 3      | Difference     | 22              | 22            | Pass   |
| 4         | 45     | 1      | Sum of squares | 41              | 41            | Pass   |
| 5         | 45     | 2      | Sum of cubes   | 189             | 189           | Pass   |

| Test Case Number | Choice | Operation | Expected Output | Actual Output | Result |
|------------------|--------|-----------|-----------------|---------------|--------|
| 6                | 45     | 3         | Difference      | 148           | 148    |
|                  |        |           |                 |               | Pass   |

## Verification

For 123:

**Sum of squares:**

$$1^2 + 2^2 + 3^2 = 1 + 4 + 9 = 14$$

**Sum of cubes:**

$$1^3 + 2^3 + 3^3 = 1 + 8 + 27 = 36$$

**Difference:**

$$36 - 14 = 22$$

## 3. C Program

```
#include <stdio.h>

int main()
{
    int num, temp, digit, choice;
    int sumSquare = 0, sumCube = 0, difference;

    printf("Enter a positive integer: ");
    scanf("%d", &num);

    printf("\nMENU\n");
    printf("1. Sum of squares of digits\n");
    printf("2. Sum of cubes of digits\n");
    printf("3. Difference between sum of cubes and squares\n");

    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch(choice)
    {
        case 1:
            temp = num;
            sumSquare = 0;

            while(temp > 0)
            {
                digit = temp % 10;
```

```

        sumSquare = sumSquare + digit * digit;
        temp = temp / 10;
    }

    printf("Sum of squares = %d\n", sumSquare);
    break;

case 2:
    sumCube = 0;

    for(temp = num; temp > 0; temp = temp / 10)
    {
        digit = temp % 10;
        sumCube = sumCube + digit * digit * digit;
    }

    printf("Sum of cubes = %d\n", sumCube);
    break;

case 3:
    temp = num;
    sumSquare = 0;
    sumCube = 0;

    do
    {
        digit = temp % 10;

        sumSquare = sumSquare + digit * digit;
        sumCube = sumCube + digit * digit * digit;

        temp = temp / 10;
    } while(temp > 0);

    difference = sumCube - sumSquare;

    printf("Difference = %d\n", difference);
    break;

default:
    printf("Invalid choice.\n");
}

return 0;
}

```

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## 4. Compile and Execute in Kali Linux

Save the program as:

diff.c

Compile it using the `cc` compiler:

```
cc diff.c
```

Execute:

```
./a.out
```

```
(kali㉿kali)-[~/Desktop]
$ cc diff.c
```

```
(kali㉿kali)-[~/Desktop]
$ ./a.out
Enter a positive integer: 123
```

MENU

1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and squares

Enter your choice: 1

Sum of squares = 14

```
(kali㉿kali)-[~/Desktop]
$ ./a.out
```

Enter a positive integer: 123

MENU

1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and squares

Enter your choice: 2

Sum of cubes = 36

```
(kali㉿kali)-[~/Desktop]
$ ./a.out
```

Enter a positive integer: 123

MENU

1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and squares

Enter your choice: 3

Difference = 22

```
(kali㉿kali)-[~/Desktop]
$ ./a.out
Enter a positive integer: 45

MENU
1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and squares
Enter your choice: 1
Sum of squares = 41

(kali㉿kali)-[~/Desktop]
$ ./a.out
Enter a positive integer: 45

MENU
1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and squares
Enter your choice: 2
Sum of cubes = 189
```

```
(kali㉿kali)-[~/Desktop]
$ ./a.out
Enter a positive integer: 45

MENU
1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and squares
Enter your choice: 3
Difference = 148
```

---

## 5. Testing Against Prepared Use Cases

### Test Case 1 — Sum of Squares

#### Input:

Number = 123  
Choice = 1

#### Calculation:

$$1^2 + 2^2 + 3^2 = 14$$

**Expected Output:** 14

**Actual Output:** 14

**Result:** PASS ✓

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## Test Case 2 — Sum of Cubes

### Input:

Number = 123

Choice = 2

### Calculation:

$$1^3 + 2^3 + 3^3 = 36$$

**Expected Output:** 36

**Actual Output:** 36

**Result:** PASS ✓

---

## Test Case 3 — Difference

### Input:

Number = 123

Choice = 3

Sum of squares = 14

Sum of cubes = 36

$$\text{Difference} = 36 - 14 = 22$$

**Expected Output:** 22

**Actual Output:** 22

**Result:** PASS ✓

## Result

The menu-driven C program was successfully compiled and executed. The **switch-case**, **while**, **for**, and **do-while** statements were implemented as required, and the program produced the correct results for the prepared test cases.